STRATEGIC PLANNING REPORT

Project-Title: Data-Driven Inventory Management and Delivery Optimization for an E-Commerce Logistics Company

Project Area: Logistics Data Analysis
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Project Title: Strategic Planning and Data Exploration for E-Commerce Logistics

WEEK 2 - Data Collection, Cleaning and Preprocessing for E-Commerce Logistics

2.1. Introduction:

Data quality is essential for accurate logistics analysis. E-commerce datasets may contain missing values, duplicate records, incorrect dates, invalid quantities, and unusual shipping costs.

Week 2 focuses on preparing the dataset for analysis and machine learning.

2.2. Dataset Characteristics / Dataset Description:

The proposed dataset contains approximately 5,000–10,000 e-commerce order records.

The main variables include:

- Order ID
- Order date

- Product ID
- Product category
- Quantity
- Price
- Warehouse
- Customer city
- Shipping mode
- Shipping cost
- Inventory level
- Reorder level
- Delivery days
- Order status
- Customer rating

2.3. Load the Dataset / Data Collection

```
import pandas as pd
df = pd.read_csv("ecommerce_logistics.csv")
print(df.head())
print(df.shape)
print(df.info())
```

Output:

print(df.head())

Order_ID	Order_Date	Customer_ID	Product_Category
ORD1001	2026-01-02	CUST001	Electronics
ORD1002	2026-01-03	CUST002	Fashion
ORD1003	2026-01-04	CUST003	Home & Kitchen
ORD1004	2026-01-05	CUST004	Beauty
ORD1005	2026-01-06	CUST005	Electronics

print(df.shape()) – (30,13)

ecommerce_logistics.csv contains 30 rows and 13 columns

print(df.info()) –information about dataset like data type, rows, clumns…

2.4. Data Cleaning

A Sample Chart showing the **Number of data-quality issues before and after cleaning**.

Data Issue	Before Cleaning
Missing Values	250
Duplicate Records	120
Invalid Quantities	45
Outliers	85

The chart presents the major data-quality issues identified in the e-commerce logistics dataset. Missing values, duplicate records, invalid values, and outliers can affect the accuracy of subsequent analysis and therefore need to be addressed during the data-cleaning stage.

Data Preprocessing Flow Diagram:

Raw Data
↓
Missing Value Detection
↓
Duplicate Removal
↓
Outlier Detection
↓
Invalid Data Correction
↓
Feature Engineering
↓
Normalization
↓
Clean Dataset

2.5. Check Missing Values

```
missing_values = df.isnull().sum()
print(missing_values)
```

Calculate missing percentages:

```
missing_percentage = (
   df.isnull().sum() / len(df)
) * 100
print(missing_percentage)
```

Output: ecommerce_logistics.csv this data 0% missing values in every column have data no empty.

2.5.1 Handle Missing Values

For numerical variables:

```
df["Inventory_Level"] = df["Inventory_Level"].fillna(
```

```
    df["Inventory_Level"].median())
For categorical variables:
df["Shipping_Mode"] = df["Shipping_Mode"].fillna(
    df["Shipping_Mode"].mode()[0])
```

Output:

```
In case dataset have Missing Value:
Inventory_Level     Shipping_Cost   Shipping_Mode
100          650      Express
NaN          180      Standard
80  NaN      Express
120          320      NaN
After checking the missing value:
Inventory_Level     Shipping_Cost   Shipping_Mode
100          650      Express
100          180      Standard
80  320      Express
120          320      Express
```

2.5.2. Remove Duplicate Records

```
print("Duplicates:", df.duplicated().sum())
df = df.drop_duplicates()
```

output:

```
in case dataset had:
ORD1001 ... Electronics ... 45000
ORD1002 ... Fashion ... 2500
ORD1002 ... Fashion ... 2500
Duplicates: 1
Before: 30 rows
Duplicates: 1
df = df.drop_duplicates()
After: 29 rows
```

2.5.3. Check Invalid Values

```
print(df[df["Quantity"] <= 0]) //Empty index
print(df[df["Shipping_Cost"] < 0]) // shipping costs below zero and -ve
print(df[df["Delivery_Days"] < 0]) //delivery time in negative
```

Invalid records should be investigated before removal.

2.5.4. Date Conversion

```
df["Order_Date"] = pd.to_datetime(
df["Order_Date"])
```

Create useful time features:

```
df["Year"] = df["Order_Date"].dt.year
df["Month"] = df["Order_Date"].dt.month
df["Day_of_Week"] = df["Order_Date"].dt.day_name()
```

2.6. Outlier Detection

```
Q1 = df["Shipping_Cost"].quantile(0.25)
Q3 = df["Shipping_Cost"].quantile(0.75)
```

IQR (Interquartile Range) = Q3 - Q1

lower = Q1 - 1.5 * IQR

upper = Q3 + 1.5 * IQR

```
outliers = df[(df["Shipping_Cost"] < lower) |(df["Shipping_Cost"] >
upper)]
print(outliers)
```

Outliers should be investigated rather than automatically deleted.

2.7. Feature Engineering

Total_Sales = Quantity × Unit_Price

Calculate total sales:

```
df["Total_Sales"] = (df["Quantity"] * df["Unit_Price"])
```

Create inventory status:

```
df["Inventory_Status"] = df.apply(lambda row:
  "Reorder Required"
  if row["Inventory_Level"] <= row["Reorder_Level"]
```

```
else "Sufficient",
axis=1) // row by row
```

Output:

Suppose Our data had:

Product	Inventory_Level	Reorder_Level	Inventory_Status
Electronics	20	30	Reorder Required
Fashion	50	30	Sufficient
Beauty	15	15	Reorder Required
Grocery	40	25	Sufficient

2.8. Normalization

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
columns = ["Quantity", "Inventory_Level",
"Shipping_Cost","Delivery_Days"]
df[columns] = scaler.fit_transform(df[columns])
```

Ouput:

Order	Inventory_Level	Shipping_Cost	Delivery_Days
ORD1001	100	650	3
ORD1002	80	180	2
ORD1003	60	320	5
ORD1004	120	120	2

These values are on different scales.

- Inventory Level: **60–120**
- Shipping Cost: **120–650**
- Delivery Days: **2–5**

Standard Scaler converts them to standardized values.

Before standardization

Inventory_Level	Shipping_Cost	Delivery_Days
100	650	3
80	180	2
60	320	5
120	120	2
90	550	4

After standardization

The numbers will become values roughly like:

Standardized value = (Original value − Mean) / Standard deviation

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2.9. Preprocessing Pipeline

Raw E-Commerce Data
↓
Missing Values
↓
Duplicate Removal
↓
Invalid Values
↓
Outlier Detection
↓
Date Conversion
↓
Feature Engineering
↓
Categorical Encoding
↓
Normalization
↓
Clean Dataset

2.10. Conclusion

The preprocessing stage ensures that the dataset is accurate, consistent, and suitable for statistical analysis and machine learning.